THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 4: **Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which compound cannot react with sodium metal? 1

(a) ethanoic acid **(b)** ethanol **(c)** ethane **(d)** water

2. Assertion (A): Methane gives substitution reactions with chlorine. Reason (R): Methane is an unsaturated hydrocarbon. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. The number of hydrogen molecules that add to one molecule of ethyne to convert it into ethane is: 1

(a) 1 **(b)** 3 **(c)** 2 **(d)** 4

4. Among ethane, ethene and ethyne, which has the shortest C-C bond? 1

(a) ethane **(b)** ethene **(c)** ethyne **(d)** all equal

5. Assertion (A): Esters have a sweet smell. Reason (R): Esters are formed from carboxylic acid and alcohol. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. The number of structural isomers of C_5H_{12} is: 1

(a) 2 **(b)** 4 **(c)** 5 **(d)** 3

7. Alkaline $KMnO_4$ is added drop by drop to warm ethanol. The purple colour disappears because: 1

(a) ethanol is oxidised and $KMnO_4$ is reduced
(b) water is added
(c) ethanol is reduced
(d) no reaction occurs

8. Assertion (A): Alkyl groups attached to a functional group change physical properties but not chemical properties. Reason (R): Chemical properties are determined by the functional group. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. **Assertion (A):** Covalent compounds generally have low melting and boiling points. 1
Reason (R): The intermolecular forces in covalent compounds are weak.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Esters have a sweet, fruity smell. 1
Reason (R): Esterification is the reaction of a carboxylic acid with an alcohol in the presence of an acid catalyst.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Identify the gas evolved when ethanol reacts with sodium and state the test. 2

12. Account for the fact that ethanol dissolves in water while hexane does not. 2

13. Why can we not make soap from mineral oil? 2

14. Which has the higher boiling point: CH_3CH_2OH or CH_3OCH_3? Explain. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Why does the addition of a few drops of acid not affect the rate of saponification but alkali does? 3

16. Why does carbon not form C^{4+} or C^{4-} ions, and what does it do instead? 3

17. Write the equations for the preparation of soap from a fat and the effect of adding common salt to precipitate it. 3

OR

Soap solution is added to hard water and to distilled water. Predict the observation with reasons. 3

18. Explain the importance of the functional group with reference to ethanol and ethanoic acid properties. 3

OR

Why does ethanoic acid have higher boiling point than ethanol? Give the boiling points qualitatively. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Rohan washed a white shirt with bar soap in hard water and noticed a sticky white scum on the shirt. His mother then washed it with a detergent and it was cleaned without any scum. The teacher explained that soap molecules form spherical clusters in water that trap dirt. 5
(i) What is the white scum?
(ii) Which part of a soap molecule is hydrophobic?
(iii) Name the clusters formed by soap molecules in water.
(iv) Why do detergents work in hard water while soaps do not?

20. A hydrocarbon X burns with a sooty flame and adds 1 mol of hydrogen in the presence of Ni to give a compound that burns with a clean flame. If X has 2 C atoms, identify X, write both equations and the product's name. 5

OR

A student has ethanol, ethanoic acid, ethene and ethane. Plan tests with expected observations to identify each. 5

END OF QUESTION PAPER — Total: 40 Marks